

impossible to make a scientific career. (There is a distinction, however, between the mind of a pure mathematician thriving on abstraction and that of a scientist consumed by curiosity. A mathematician is absorbed in what goes into his head while a scientist searches into what is outside of himself.) However, some people's concern for randomness can be excessive; I have even seen people trained in some fields, like say, quantum mechanics, push the idea to the other extreme, only seeing alternative histories and ignoring the one that actually took place.

Some traders can be unexpectedly introspective about randomness. I recently had dinner at the bar of the Odeon with Lauren R., a trader who was reading a draft of this book. We flipped a coin to see who was going to pay for the meal. I lost and paid. He was about to thank me when he abruptly stopped and said: "Reading your book you would say that I paid for half of it probabilistically".

I thus view people distributed across two polar categories: on one extreme, those who never accept the notion of randomness; on the other, those who are tortured by it. When I started on Wall Street in the 1980s, trading rooms were populated with people with a "business orientation", that is, generally devoid of any introspection, flat as a pancake, and likely to be fooled by randomness. Their failure rate was extremely high, particularly when financial instruments gained in complexity. Somehow tricky products, like exotic options, were introduced and carried counterintuitive payoffs that were too difficult for someone of such culture to handle. They dropped like flies; I do not think that many of the hundreds of MBAs of my generation I met on Wall Street in the 1980s still engage in such forms of professional and disciplined risk taking.

#### SALVATION VIA AEROFLOT

The 1990s witnessed the arrival of people of richer and more interesting backgrounds, which made the trading rooms far more entertaining. I was saved from the conversation of MBAs. Many scientists, some of them extremely successful in their field, arrived with a desire to make a buck. They, in turn, hired people who resembled them. While most of these people were not Ph.D.s (indeed, the Ph.D. is still a minority), the

culture and values suddenly changed, becoming more tolerant of intellectual depth. It caused an increase in the already high demand for scientists on Wall Street, owing to the rapid development of financial instruments. The dominant specialty was physics, but one could find all manner of quantitative backgrounds among them. Russian, French, Chinese, and Indian accents (by order) began dominating in both New York and London. It was said that every plane from Moscow had at least its back row full of Russian mathematical physicists *en route* to Wall Street (they lacked the street smarts to get good seats). One could hire very cheap labor by going to JFK airport with a (mandatory) translator, randomly interviewing those that fitted the stereotype. Indeed, by the late 1990s one could get someone trained by a world-class scientist for almost half the price of an MBA. As they say, marketing is everything; these guys do not know how to sell themselves.

I had a strong bias in favor of Russian scientists; many can be put to active use as chess coaches (I also got a piano teacher out of the process). In addition, they are extremely helpful in the interview process. When MBAs apply for trading positions, they frequently boast "advanced" chess skills on their résumés. I recall the MBA career counselor at Wharton recommending our advertising chess skills "because it sounds intelligent and strategic". MBAs, typically, can interpret their superficial knowledge of the rules of the game into "expertise". We used to verify the accuracy of claims of chess expertise (and the character of the applicant) by pulling a chess set out of a drawer and telling the student, now turning pale: "Yuri will have a word with you".

The failure rate of these scientists, though, was better, but only slightly so than that of MBAs; but it came from another reason, linked to their being on average (but only on average) devoid of the smallest bit of practical intelligence. Some successful scientists had the judgment (and social graces) of a door knob – but by no means all of them. Many people were capable of the most complex calculations with utmost rigor when it came to equations, but were totally incapable of solving a problem with the smallest connection to reality; it was as if they understood the letter but not the spirit of the math. I am convinced that X, a likeable Russian man of my acquaintance, has two brains: one for math and another, considerably inferior one, for everything else (which included solving problems related to the mathematics of finance). But on